

Haoran (Cris) Wang

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EDUCATION

Johns Hopkins University — M.S. Mechanical Engineering

Jan. 2026–Present

Southern University of Science and Technology (SUSTech) — B.Eng. Robotics Engineering

Sep. 2020–Jun. 2025

INTERNSHIPS

UBTech Robotics

Motion Control Algorithm Intern

Sep. 2025–Jan. 2026

- Independently developed a real-time multi-source motion-retargeting framework that unified motion-capture (Xsens), VR (PICO), and RGB-video inputs into a common robot-control pipeline.
- Adapted a GMR-based whole-body retargeting pipeline using **MINK differential IK**, tuning task weights and velocity/configuration regularization across motion-capture, VR, and vision inputs to improve tracking stability and suppress foot penetration and sliding during humanoid teleoperation.
- Deployed the system for real-time teleoperation on the Walker S2 humanoid with motion-capture and PICO VR inputs at **~300 ms** end-to-end latency, and extended it to teleoperation-based data collection for UBTECH humanoids and wheeled dual-arm robots.

RESEARCH EXPERIENCES

Multimedia Laboratory (HKU)

Research Assistant

Mar. 2026–Sep. 2026

- Independently designed and built a wearable egocentric sensing prototype from mechanical structure to embedded integration, combining nine fisheye cameras, three IMUs, wrist-mounted sensing modules, AprilTag-based localization, and hardware-level synchronization.
- Built a real-world data pipeline and collected ~10 hours of synchronized motion data; developed a custom ground-truth pipeline combining external multi-view cameras with optical-mocap rigid-body tracking to recover a scale-consistent 3D human skeleton.
- Re-implemented and fine-tuned EgoRear's three-stage heatmap-based 3D pose-estimation pipeline, benchmarking against EgoPoseFormer and other egocentric methods.
- Achieved **50.5 mm MPJPE** on a held-out real-world test set versus **56.94 mm** for the EgoRear baseline (**~11% lower error**), using ~10 hours of real-world data without large-scale synthetic-data pretraining.

Human-Augmented Robotics Lab (SUSTech)

Undergraduate Researcher

Jan. 2024–Mar. 2025

- Developed a wearable gait-sensing system integrating 7 IMUs and an 18-channel resistive pressure insole, with synchronized treadmill ground-reaction-force measurements as reference.
- Predicted a 100-point vGRF curve from IMU data with a lightweight FNN, then optimized **gait-phase-specific** calibration weights for the 18 insole sensors offline; reduced vGRF RMSE **from 24.7% to 10.2%** of body weight, with similar accuracy under variable-speed walking.
- H. Wang, T. Su, C. Fu, and Y. Leng, "An Off-line Self-calibration Method for Resistive Insole Pressure Sensors," RCAR, 2025.

Robot Mobility & Manipulation Lab (SUSTech)

Undergraduate Researcher

Feb. 2023–Mar. 2024

- Independently designed and built SWhegPro3, a four-wheel-drive stair-climbing robot with three-impeller wheel-leg modules, self-locking push rods, brushless motor drives, and ROS/CAN/PWM control.
- Used MATLAB kinematic models to optimize wheelbase and map stair dimensions to push-rod extension, adjusting different stairs.
- Used speed-loop control with separate front/rear wheel torque limits after position-dependent phase tracking proved unstable; validated 1.8 steps/s peak climbing speed and a 7 kg payload in field tests.
- H. Wang, S. Wang, C. Dai, and Z. Jia, "SWhegPro3: A Three-Impeller Wheel-Leg Transformable Robot with Variable Robust Adaptability to Stair Dimensions," *ROBIO*, 2023.
- H. Wang et al., "Ubiquitous Field Transportation Robots with Robust Wheel-Leg Transformable Modules," arXiv:2410.18507, 2024.

COURSE PROJECT

Persistent Object-Centric Memory for Embodied AI

Feb. 2026–Mar. 2026

- Co-developed an RGB-D object segmentation pipeline using RANSAC plane removal, point-cloud clustering, and SAM2 mask refinement.
- Initialized object-centric scene memory with class labels, masks, bounding boxes, and world-frame 3D centroids for detected objects.

SKILLS

Programming & AI Tools: Python, MATLAB, C/C++, Java; Codex, Cursor

Deep Learning: PyTorch, TensorFlow, CNNs, Transformers

Computer Vision: Egocentric vision, image classification, localization, segmentation, point-cloud processing, camera calibration, OpenCV

Robotics: ROS, Gazebo, MuJoCo, Isaac Sim, BlenderProc, motion retargeting, sensor fusion

Hardware & Sensing: SolidWorks, Keil, AutoCAD, motion capture (Xsens/PICO VR), pressure sensors